

ferred securely while minimizing the amount of computational power required to do the actual encryption or decryption.

Key strength

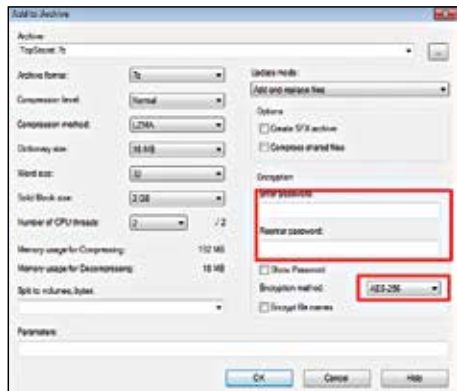
Key strength usually refers to the length of the key being used for the algorithm. However it isn't always the measure of its cryptographic strength. Randomness of characters used in the key is also an important factor in determining the strength of the cipher.

Hashing

While you can encrypt passwords for any system (website user passwords, operating systems account passwords or passwords to access a program), it's possible to decrypt them. Since passwords are still the major way to authenticate users for various purposes, they do pose a threat to security.

Make that a serious threat to security in the case of Open Source systems such as Linux. It's common knowledge that Open Source can allow hackers to know about the weaknesses or loopholes of the software more easily than Closed Source software. If the password file gets into the wrong hand, security would be a thing of the past. Obviously, storing passwords in plaintext (original) format would be downright stupid!

For these reasons passwords are neither stored in encrypted form nor in plain format. So how are they stored? Well, they're stored in 'hashed' form. Hashing is a technique to map text of variable length to a string of fixed length in such a way that the only way to find out the original text from the hash is to run a brute force attack. Since hashing functions are irreversible, they don't qualify strictly as cryptographic functions. However, they're associated with security often enough and it's easy to find them mentioned alongside passwords in the context of cryptography.



Compression software such as 7-zip help in encrypting compressed archives